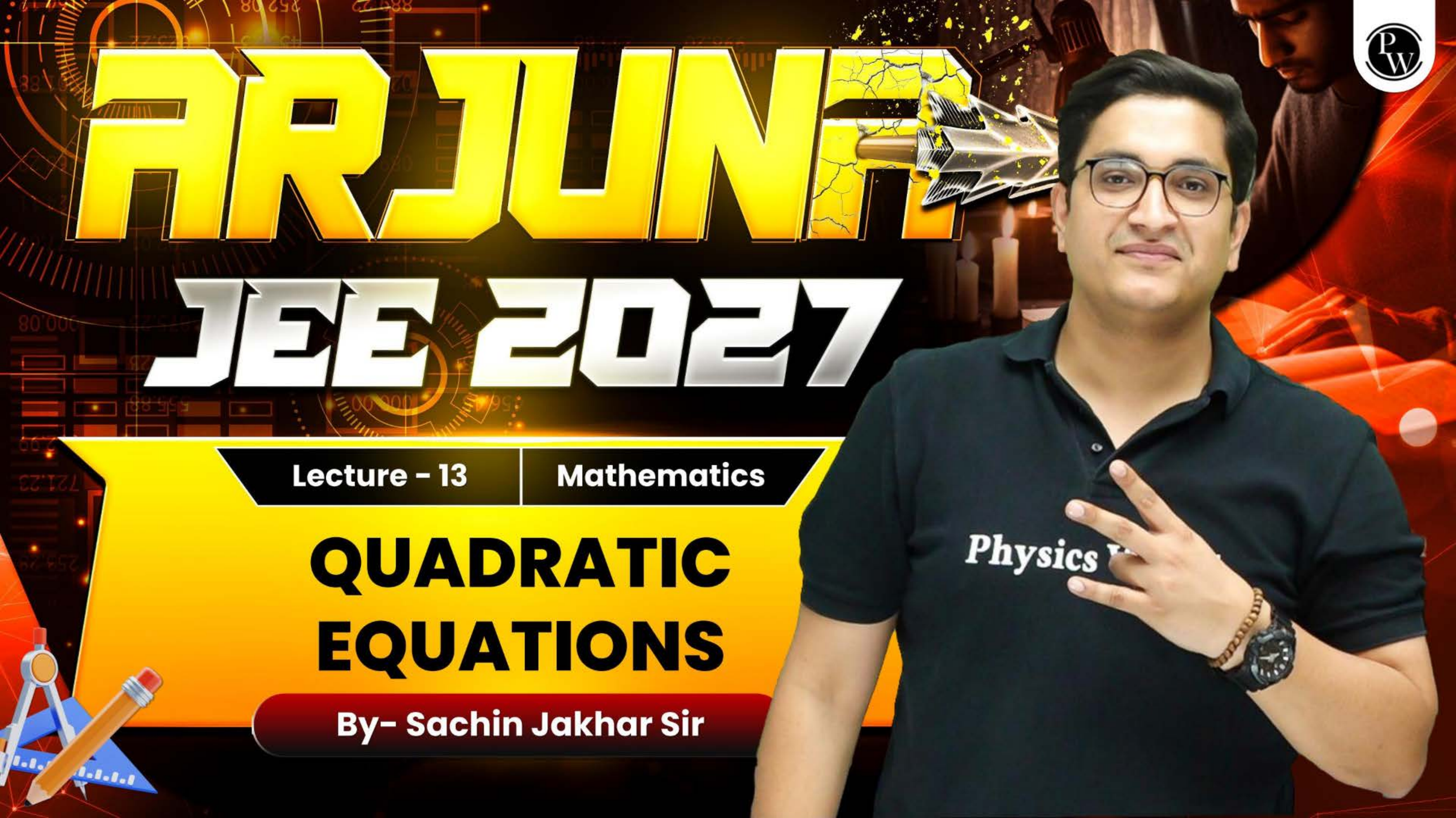
ARJUNA JEE 2027

Lecture - 13 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Theory of Higher Degree Equations

B Transformation of Equation

C General Second Degree Equation

D Question Practice



Question Practice

Acidic Question

If the equations $x^3 + x^2 - 4x = 4$ and $x^2 + px + 2p = 0\ (p \in R)$ have two roots common, then the value of p is

DIBY-A

A -2

B -1

C 1

D 3

Acidic Question

The values of k, for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ posses atleast one positive root, are

DIBY-B

A $[4, \infty)$

B $(-\infty, -1) \cup [4, \infty)$

C $[-1, 4]$

D $(-\infty, -1]$

Challenger Question

Q_1

For what real values of a do the roots $x^2 - 2x - a^2 + 1 = 0$ lie between the roots $x^2 - 2(a+1)x + a(a-1) = 0$.

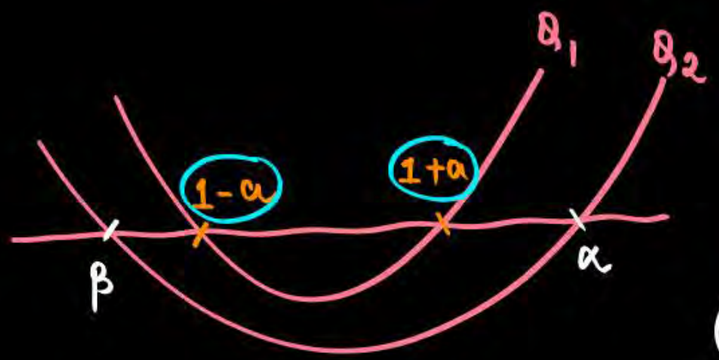
Q_2

α, β

$x^2 - 2x + (1 - a^2) = 0$

$1+a$, $1-a$

$(1-a)$ $(1+a)$

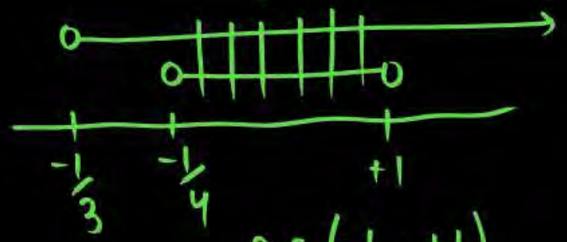


$af(1+a) < 0 \cap a f(1-a) < 0$

$1 \cdot \left((a+1)^2 - 2(a+1)(a+1) + a(a-1)\right) < 0$

$(1-a)^2 - 2(1+a)(1-a) - a(1-a) < 0$

$f \equiv f$



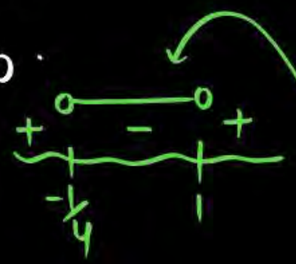
$a \in \left(-\frac{1}{4}, +1\right)$

$-(a+1)^2 + a^2 - a < 0$

$-a^2 - 2a - 1 + a^2 - a < 0$

$-1 < 3a$

$-\frac{1}{3} < a$



$(1-a)\{1-a-2-2a-a\} < 0$

$(1-a)(-1-4a) < 0$

$(a-1)(4a+1) < 0$

Challenger Question



Find the values of m for which the expression: $\frac{2x^2 - 5x + 3}{4x - m}$ can take all real values for $x \in R - \left\{\frac{m}{4}\right\}$

[Ans. $m \in (4, 6)$]

Range $\in R$

$$y = \frac{2x^2 - 5x + 3}{4x - m}$$

$$4xy - my = 2x^2 - 5x + 3$$

$$2x^2 - (5 + 4y)x + 3 + my = 0$$

$D \geq 0$

$$(5+4y)^2 - 4 \times 2 \times (3 + my) \geq 0$$

$$25 + 16y^2 + 40y - 24 - 8my \geq 0$$

$$16y^2 + 4(10 - 2m)y + 1 \geq 0$$

$16y^2 + 4(10-2m)y + 1 \geq 0 \longrightarrow y \in R$

$a > 0$ ⟶ $16 > 0$ ✓

$D \leq 0$

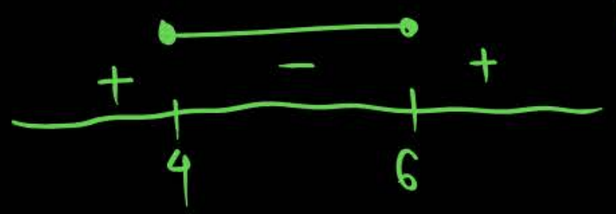
$16(10-2m)^2 - 4 \times 16 \times 1 \leq 0$

$(5-m)^2 - 1 \leq 0$

$m^2 + 25 - 10m - 1 \leq 0$

$m^2 - 10m + 24 \leq 0$

$(m-6)(m-4) \leq 0$



$m \in [4,6]$

$m \in (4,6)$ fnf

Trick!

always Check end points

$m = 4, 6$.

$$y = \frac{2x^2 - 5x + 3}{4x - m}$$

common factor → $\frac{Q}{L}$ → L

N^r: $2x^2 - 5x + 3 = 0$ → $x = 1$, $x = 3/2$

D^r: $4x - m = 0$

$x = 1$ ↓ → $4 = m$ → X

$x = 3/2$ → $m = 6$ → X

explanation

if $m = 4$ ×

$$y = \frac{2x^2 - 5x + 3}{4x - 4} = \frac{(x-1)(2x-3)}{4(x-1)}$$

$$y = \frac{2x-3}{4}, \quad x \neq 1$$

$x = 1$

Range → $R - \left\{\frac{-1}{4}\right\}$



RATIONAL ROOT THEOREM

Lec-12,13 (Basic Mathematics)

Factorise following polynomial equations:

1. **Cubic :** $x^3 - 5x^2 - 2x + 6 = 0$
2. **Cubic :** $6x^3 + 35x^2 + 48x + 16 = 0$
3. **Quartic :** $x^4 + 11x^3 + 41x^2 + 61x + 30 = 0$



ANSWERS

1. $1, 2+\sqrt{10}, \ 2-\sqrt{10}$
2. $-4, -\frac{1}{2}, -\frac{4}{3}$
3. $-1, -2, -3, -5$

Bumper + Challenger Question

If α, β, γ are roots of cubic : $x^3 - x^2 + 1 = 0$, then find value of:

(i) $\sum\alpha = \alpha+\beta+\gamma = 1$ (ii) $\sum\alpha\beta = \alpha\beta+\beta\gamma+\gamma\alpha = 0$ (iii) $\alpha\beta\gamma = -1$

(iv) $\sum\alpha^2 = \alpha^2+\beta^2+\gamma^2 = 1$ (v) $\sum\alpha^3 = \alpha^3+\beta^3+\gamma^3 = -2$ (vi) $\sum\frac{1}{\alpha^2}$

(vii) $\sum\alpha^2\beta$ (viii) $\sum\alpha^5$ (ix) $\sum\frac{1}{\alpha^3+1}$

$$\overset{+}{1}x^3 \overset{-}{-} x^2 \overset{+}{+} 0x \overset{-}{+} 1 = 0$$

$$\alpha+\beta+\gamma = \frac{-(-1)}{1} = 1$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = 0$$

$$\alpha\beta\gamma = -1$$

$\alpha+\beta+\gamma = 1$

$\alpha\beta+\beta\gamma+\gamma\alpha = 0$

$\alpha\beta\gamma = -1$

(iv) SBS:

$$\underbrace{\alpha^2+\beta^2+\gamma^2}_{} + 2\underbrace{(\alpha\beta+\beta\gamma+\gamma\alpha)}_{0} = 1$$

(v)

$$\underbrace{\alpha^3+\beta^3+\gamma^3}_{} - 3\overbrace{\alpha\beta\gamma}^{-1}$$

$$= \underbrace{(\alpha+\beta+\gamma)}_{1}\left(\underbrace{\alpha^2+\beta^2+\gamma^2}_{1} - \underbrace{\alpha\beta-\beta\gamma-\gamma\alpha}_{0}\right)$$

$$\alpha^3+\beta^3+\gamma^3 - 3(-1) = 1\times(1-0)$$

$$\alpha^3+\beta^3+\gamma^3 = 1-3$$

$$= -2$$

Bumper + Challenger Question

If α, β, γ are roots of cubic : $x^3 - x^2 + 1 = 0$, then find value of:

(i) $\Sigma\alpha = \alpha+\beta+\gamma = 1$ (ii) $\Sigma\alpha\beta = \alpha\beta+\beta\gamma+\gamma\alpha = 0$ (iii) $\alpha\beta\gamma = -1$

(iv) $\Sigma\alpha^2 = \alpha^2+\beta^2+\gamma^2 = 1$ (v) $\Sigma\alpha^3 = \alpha^3+\beta^3+\gamma^3 = -2$ (vi) $\Sigma\frac{1}{\alpha^2} = \frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{1}{\gamma^2} = 2$.

(vii) $\Sigma\alpha^2\beta = 3$. (viii) $\Sigma\alpha^5 = \alpha^5+\beta^5+\gamma^5 = -4$ (ix) $\Sigma\frac{1}{\alpha^3+1}$

****

(V) (M-2) C.T.

$x^3 - x^2 + 1 = 0$ — α, β, γ

(vii) $\Sigma\alpha^2\beta = \alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta = 3$.

$\alpha^3 - \alpha^2 + 1 = 0$

$\alpha^3 = \alpha^2 - 1$

$\beta^3 = \beta^2 - 1$

$\gamma^3 = \gamma^2 - 1$

add

$\alpha^3+\beta^3+\gamma^3 = \alpha^2+\beta^2+\gamma^2 - 3 = 1-3 = -2$

$\times\alpha^2$

$\alpha^5 = \alpha^4 - \alpha^2$

$\beta^5 = \beta^4 - \beta^2$

$\gamma^5 = \gamma^4 - \gamma^2$

add

$\alpha^5+\beta^5+\gamma^5 = -3 - (1) = -4$

$\times\alpha$

$\alpha^4 = \alpha^3 - \alpha$

$\beta^4 = \beta^3 - \beta$

$\gamma^4 = \gamma^3 - \gamma$

add

$\alpha^4+\beta^4+\gamma^4 = (\alpha^3+\beta^3+\gamma^3) - (\alpha+\beta+\gamma)$

$= -2 - 1 = -3$

$$x^3 - x^2 + 0x + 1 = 0$$

$$\alpha+\beta+\gamma = \frac{-(-1)}{1} = 1$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = 0$$

$$\alpha\beta\gamma = -1$$

(vii)

$$S_1 \times S_2 = (\alpha+\beta+\gamma)(\alpha\beta+\beta\gamma+\gamma\alpha)$$

$$= \alpha^2\beta + \alpha\beta\gamma + \gamma\alpha^2 + \alpha\beta^2 + \beta^2\gamma + \alpha\beta\gamma + \alpha\beta\gamma + \beta\gamma^2 + \gamma^2\alpha$$

$$1 \times 0 = 3(\alpha\beta\gamma) + \left(\sum \alpha^2\beta\right)$$

($\alpha\beta\gamma = -1$)

$$\sum \alpha^2\beta = 3.$$

(vi)

$$\# \quad \alpha\beta+\beta\gamma+\gamma\alpha = 0$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2(\alpha\beta^2\gamma + \beta\gamma^2\alpha + \alpha^2\beta\gamma) = 0$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\beta+\gamma+\alpha) = 0$$

($\alpha\beta\gamma = -1$, $\beta+\gamma+\alpha = 1$)

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = 2$$

Bumper + Challenger Question

If α, β, γ are roots of cubic : $x^3 - x^2 + 1 = 0$, then find value of:

(i) $\Sigma\alpha$ (ii) $\Sigma\alpha\beta$ (iii) $\alpha\beta\gamma$

(iv) $\Sigma\alpha^2$ (v) $\Sigma\alpha^3$ (vi) $\Sigma\frac{1}{\alpha^2}$

(vii) $\Sigma\alpha^2\beta$ (viii) $\Sigma\alpha^5$ ~~(ix)~~ $\Sigma\frac{1}{\alpha^3+1}$



$\alpha^3 + 1 = \alpha^2$

$\frac{1}{\alpha^3+1} = \frac{1}{\alpha^2}$

$= \frac{1}{\alpha^3+1} + \frac{1}{\beta^3+1} + \frac{1}{\gamma^3+1}$

$= \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

$= \Sigma\frac{1}{\alpha^2}$

$= (vi)$

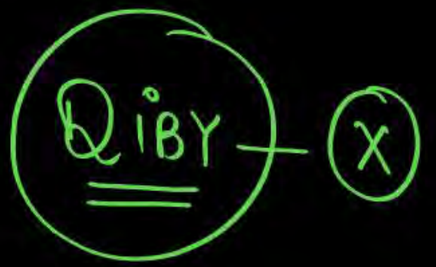
Bumper + Challenger Question

If α, β, γ are roots of cubic : $x^3 - 3x^2 - 4x + 1 = 0$, then find value of:

(i) $\Sigma\alpha$ (ii) $\Sigma\alpha\beta$ (iii) $\alpha\beta\gamma$

(iv) $\Sigma\alpha^2$ (v) $\Sigma\alpha^3$ (vi) $\Sigma\frac{1}{\alpha^2}$

(vii) $\Sigma\alpha^2\beta$ (viii) $\Sigma\alpha^5$ (ix) $\Sigma\frac{1}{\alpha^3+1}$



$S_1 = \alpha + \beta + \gamma = -\frac{(-1)}{1} = 1$

$\alpha\beta + \beta\gamma + \gamma\alpha = 0$

$\alpha\beta\gamma = -1$

Ques:-

$x^3 - x^2 + 0x + 1 = 0$ $\quad \alpha, \beta, \gamma$

$S_n = \alpha^n + \beta^n + \gamma^n$

$S_5 = \alpha^5 + \beta^5 + \gamma^5 = ??$

$S_n - S_{n-1} + 0S_{n-2} + S_{n-3} = 0$

$S_n - S_{n-1} + S_{n-3} = 0$

$n=5$: $S_5 - S_4 + S_2 = 0$

$S_5 = S_4 - S_2$

$= -3 - 1 = -4$

$n=4$: $S_4 - S_3 + S_1 = 0$

$S_4 = S_3 - S_1$

$= -2 - 1$

$= -3$

Acidic Question

If α, β and γ are roots of cubic equation $x^3 + 3x - 1 = 0$ then find value of:

(i) $(2-\alpha)(2-\beta)(2-\gamma) = 13$.

(ii) $(3+\alpha)(3+\beta)(3+\gamma) = 37$.

(iii) $(4-\alpha^2)(4-\beta^2)(4-\gamma^2) = 195$.

(iv) $(1+\alpha^2)(1+\beta^2)(1+\gamma^2) = 5$.

α β γ

$x^3 + 3x - 1 = 1(x-\alpha)(x-\beta)(x-\gamma)$ (id.)

$x = 2$

$13 = 2^3 + 3\times 2 - 1 = (2-\alpha)(2-\beta)(2-\gamma)$

$x = -3$

$(-3)^3 + 3(-3) - 1 = (-3-\alpha)(-3-\beta)(-3-\gamma)$

$-37 = -27 - 9 - 1 = -(3+\alpha)(3+\beta)(3+\gamma)$

$(2-\alpha)(2+\alpha)(2-\beta)(2+\beta)(2-\gamma)(2+\gamma)$

13×15

M-2

$1 + (\alpha^2+\beta^2+\gamma^2) + (\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2) + \alpha^2\beta^2\gamma^2 = ??$

$i = \sqrt{-1}$ (imag.)

$i^2 = -1 \xrightarrow{SBS} i^4 = 1$

$i \times$ → $i^3 = -i$

iv

$$(i-\alpha)(i+\alpha) = (i)^2 - (\alpha)^2$$
$$= -1 - \alpha^2$$
$$= -(1+\alpha^2)$$

$$x^3 + 3x - 1 = 1(x-\alpha)(x-\beta)(x-\gamma) \quad (id.)$$

$x = i$

$$i^3 + 3i - 1 = (i-\alpha)(i-\beta)(i-\gamma)$$
$$2i - 1 = (i-\alpha)(i-\beta)(i-\gamma) \quad \text{— A}$$

$x = -i$

$$(-i)^3 + 3(-i) - 1 = (-i-\alpha)(-i-\beta)(-i-\gamma)$$
$$+(2i+1) = +(i+\alpha)(i+\beta)(i+\gamma) \quad \text{— B}$$

$$(2i-1)(2i+1) = (2i)^2 - 1^2$$
$$= 4i^2 - 1$$
$$= 4(-1) - 1$$
$$= -5$$

A × B

$$+5 = (2i-1)(2i+1) = +(1+\alpha^2) \times +(1+\beta^2) \times +(1+\gamma^2)$$

Question

If α, β, γ are roots of equation $x^3 - 2x^2 - 1 = 0$ and $T_n = \alpha^n + \beta^n + \gamma^n$, then value of $\frac{T_{11} - T_8}{T_{10}}$ is equal to

A 1

B 2

C –1

D 3

$x^3 - 2x^2 + 0x - 1 = 0$ (α, β, γ)

$T_n - 2T_{n-1} + 0\,T_{n-2} - T_{n-3} = 0$

$n = 11$

$T_{11} - 2T_{10} - T_8 = 0$

$\frac{T_{11} - T_8}{T_{10}} = 2$

THEORY OF HIGHER DEGREE EQUATIONS

Quartic: $ax^4 + bx^3 + cx^2 + dx + e = 0, \ (a \neq 0)$



S_1 : Sum of roots = $\sum\alpha = \alpha + \beta + \gamma + \delta = \frac{-b}{a}$

S_2 : Sum of product of roots taken two at a time = $\sum\alpha\beta$

$= \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{+c}{a}$

S_3 : Sum of product of roots taken 3 at a time = $\sum\alpha\beta\gamma$

$= \alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta = \frac{-d}{a}$

S_4 : Product of roots = $\alpha\beta\gamma\delta = \frac{+e}{a}$

Example

$5x^4 - 3x^3 + x - 11 = 0$

$$+ \quad - \quad + \quad - \quad +$$
$$5x^4 - 3x^3 + 0x^2 + 1x - 11 = 0$$

$S_1 = \sum\alpha = \frac{-(-3)}{5} = \frac{3}{5}$

$S_2 = \sum\alpha\beta = \frac{+0}{5} = 0$

$S_3 = \sum\alpha\beta\gamma = \frac{-1}{5} = -\frac{1}{5}$

$S_4 = \alpha\beta\gamma\delta = \frac{+(-11)}{5} = -\frac{11}{5}$



FOR ANY POLYNOMIAL EQUATION

In any polynomial equation (degree $= n$) :

S_1 = Sum of roots taken one at a time $= \dfrac{-\text{coeff. of } x^{n-1}}{\text{coeff. of } x^n} = \dfrac{-a_{n-1}}{a_n}$

S_2 = Sum of product roots taken two at a time $= \dfrac{+\text{coeff. of } x^{n-2}}{\text{coeff. of } x^n} = \dfrac{+a_{n-2}}{a_n}$

S_3 = Sum of product of roots taken three at a time $= -\dfrac{\text{coeff. of } x^{n-3}}{\text{coeff. of } x^n} = \dfrac{-a_{n-3}}{a_n}$

S_r = Sum of product of roots taken r at a time $= (-1)^r \dfrac{\text{coeff. of } x^{n-r}}{\text{coeff. of } x^n}$

$$\overset{+}{a_n} x^n \overset{-}{+} a_{n-1} x^{n-1} \overset{+}{+} a_{n-2} x^{n-2} \overset{-}{+} \ldots\ldots = 0$$

TRANSFORMATION OF EQUATION

QUESTION

If α and β are roots of $x^2 - 3x + 1 = 0$, then form a quadratic whose roots are $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$.

α, β

$\alpha+\beta = 3$

$\alpha\beta = 1$

M-1

$$x^2 - \left(\frac{\alpha}{\alpha+1} + \frac{\beta}{\beta+1}\right)x + \frac{\alpha\beta}{(\alpha+1)(\beta+1)} = 0$$

M-2 Transf.

y = new root.

x = old root. ✓

$$y = \frac{\text{old root}}{\text{old root} + 1} = \frac{x}{x+1}$$

$$xy + y = x$$

$$y = x - xy \Rightarrow y = x(1-y)$$

$$\frac{y}{1-y} = x$$

M-1

$$x^2 - \left(\frac{\alpha}{\alpha+1} + \frac{\beta}{\beta+1}\right)x + \frac{\alpha\beta}{(\alpha+1)(\beta+1)} = 0$$

$$x^2 - (1)x + \frac{1}{5} = 0$$

$$5x^2 - 5x + 1 = 0$$

$\alpha + \beta = 3$

$\alpha\beta = 1$

$$\left(\frac{\alpha}{\alpha+1} + \frac{\beta}{\beta+1}\right) = \frac{\alpha\beta + \alpha + \beta\alpha + \beta}{1 + \alpha + \beta + \alpha\beta}$$

$$= \frac{2 \times 1 + 3}{1 + 3 + 1} = \frac{5}{5} = 1$$

TRANSFORMATION OF EQUATION

QUESTION

If α and β are roots of $x^2 - 3x + 1 = 0$, then form a quadratic whose roots are $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$.

α, β

$x = \frac{y}{1-y}$

$$\frac{y^2}{(1-y)^2} - 3\left(\frac{y}{1-y}\right) + 1 = 0$$

$$\frac{y^2 - 3y(1-y) + (1-y)^2}{(1-y)^2} = 0$$

$$y^2 - 3y + 3y^2 + y^2 - 2y + 1 = 0$$

$$5y^2 - 5y + 1 = 0$$

Q-2 Transf.

$\breve{y}$ = new root.

x = old root. ✓

$$y = \frac{\text{old root}}{\text{old root} + 1} = \frac{x}{x+1}$$

$$xy + y = x$$

$$y = x - xy \Rightarrow y = x(1-y)$$

$$\frac{y}{1-y} = x$$

Ques: $7x^2 - 31x + 11 = 0$ $\begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix}$

Quad. $= 0$?? $\begin{matrix}\nearrow \frac{1}{\alpha} \\ \searrow \frac{1}{\beta}\end{matrix}$

Sol.

$\# \quad y = \frac{1}{x}$

$\hookrightarrow x = \frac{1}{y}$

$$7\left(\frac{1}{y^2}\right) - 31\left(\frac{1}{y}\right) + 11 = 0$$

$$\frac{7 - 31y + 11y^2}{y^2} = 0$$

$$7 - 31y + 11y^2 = 0 \quad \begin{matrix}\nearrow \frac{1}{\alpha} \\ \searrow \frac{1}{\beta}\end{matrix}$$

Ques: $7x^2 - 31x + 11 = 0$ — roots α, β

Quad. $= 0$?? — roots $\frac{2\alpha-1}{3\alpha+2}$, $\frac{2\beta-1}{3\beta+2}$

Sol.

$y = \frac{2x-1}{3x+2}$

$3xy + 2y = 2x - 1$

$2y + 1 = 2x - 3xy$

$2y + 1 = x(2 - 3y)$

$\frac{2y+1}{2-3y} = x$

Acidic Question

Old-IIT Jee.

Given that α, β, γ are all real roots of the equation $x^3 - 2007x + 2002 = 0$, then the value of $\frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} + \frac{\gamma-1}{\gamma+1}$ is equal to

Sum of roots of new cubic !!

$x^3 - 2007x + 2002 = 0$ — α, β, γ

$y = \frac{x-1}{x+1}$

$xy + y = x - 1$

$y + 1 = x(1-y)$

$\frac{1+y}{1-y} = x$

Hw

Cubic = 0 ??

$\frac{\alpha-1}{\alpha+1}$, $\frac{\beta-1}{\beta+1}$, $\frac{\gamma-1}{\gamma+1}$

$\left(\frac{1+y}{1-y}\right)^3 - 2007\left(\frac{1+y}{1-y}\right) + 2002 = 0$

↳ Sum of Roots = ?

DO IT BY YOURSELF (DIBY)

LOCATION OF ROOTS

DIBY-63

Let $x^2-(m-3)x+m=0,(m\in R)$ be a quadratic equation. Find the value of m for which

(i) both the roots are smaller than 2.

(ii) both the roots are greater than 2.

(iii) one root is smaller than 2 and the other root is greater than 2.

(iv) exactly one root lies in the interval (1, 2).

(v) both the roots lie in the interval (1, 2).

(vi) one root is greater than 2 and the other root is smaller than 1.

(vii) atleast one root lie in interval (1, 2).

(viii) atleast one root is greater than 2.

Check!

Ans. :

(i) $m\in(-\infty,1]$, (ii) $m\in[9,10)$, (iii) $m\in(10,\infty)$, (iv) $m\in(10,\infty)$,
(v) $m\in\phi$, (vi) $m\in\phi$, (vii) $m\in(10,\infty)$, (viii) $m\in[9,\infty)-\{10\}$

LOCATION OF ROOTS

DIBY-64

Find all values of a for which the equation $4x^2 - 2x + a = 0$ has two roots lie in the interval $(-1, 1)$.

[Ans. : $a \in \left(-2, \frac{1}{4}\right]$]

DIBY-65

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is [JEE Mains-2023]

(A) 2 (B) 0 (C) 1 (D) 3

[Ans. (C)]

DIBY-66

Find all the values of 'a' for which both roots of the equation $x^2 + x + a = 0$ exceed the quantity 'a'.

(A) $(-\infty, -2)$ (B) $\left(-\infty, \frac{1}{4}\right]$ (C) $(-2, 0)$ (D) $(-\infty, -2) \cup \left(0, \frac{1}{4}\right)$

[Ans. (A)]

LOCATION OF ROOTS

DIBY-67

Find the set of values of 'a' for which zeroes of the quadratic polynomial $(a^2+a+1)x^2+(a-1)x+a^2$ are located on either side of 3.

(A) $(-\infty,-2)$ (B) $(0,\infty)$ (C) ϕ (D) $\mathbb{R}$

[Ans. : (C)]

DIBY-68

Find all possible values of 'a' for which exactly one root of the quadratic equation $x^2-(a+1)x+2a=0$ lie in the interval (0, 2).

(A) $(-\infty,0)\cup(6,\infty)$ (B) $(-\infty,0]\cup(6,\infty)$

(C) $(-\infty,0)\cup[6,\infty)$ (D) $(-\infty,0]\cup[6,\infty)$

[Ans. : (B)]

HIGHER DEGREE POLYNOMIALS

DIBY-69

If α, β, γ are roots of $5x^3 - qx - 1 = 0, (q \in R)$ find the value of $\frac{\alpha^2-3}{\beta\gamma} + \frac{\beta^2-3}{\alpha\gamma} + \frac{\gamma^2-3}{\alpha\beta}$.

[Ans. : 3]

DIBY-70

Given that α, β, γ are the roots of cubic equation $x^3 - 3x^2 + 2x + \frac{2}{3} = 0$. The value of $\alpha^4 + \beta^4 + \gamma^4$ is equal to

[Ans. : 9]

DIBY-71

$P(x) = x^3 + 33x^2 + 327x + 935$. Let $P(x)$ be a polynomial as described above with $a; b; c$ the roots of $P(x)$. Find $a^2 + b^2 + c^2$ without solving $P(x) = 0$.

[Ans. : 435]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: (3 sittings of 1 HR each)

Exercise-02(Prabal): 1 to 10,13,16,18 to 25.

Exercise-03(Setu): 1 to 30 (except-13,16,17,21,25,26,29)

It is never about end result

It's always about Journey

Thank You

#futureIITians